

Eigen Values & Eigen Vectors

◆ **Formal Definition:** Let A be an $n \times n$ matrix.

- An **eigenvector** of A is a *nonzero* vector x in R^n such that $Ax = \lambda x$, for some scalar λ .
- An **eigenvalue** of A is a scalar λ such that the equation $Ax = \lambda x$ has a *nontrivial* solution.

Characteristic Equation of a Square Matrix

Let A be a square matrix and λ be some scalar then Characteristic Equation of a Square Matrix A is

$|A - \lambda I| = 0$ and $|A - \lambda I|$ is known as Characteristic polynomial.

Characteristic Equation of a Square Matrix A of order 3 can also be found using following formula

$$-\lambda^3 + S_1\lambda^2 - S_2\lambda + |A| = 0$$

Where $S_1 = \text{Trace } A$

$S_2 = \text{Sum of Minors of Diagonal Elements}$

$|A| = \text{Determinant of } A$

Cayley-Hamilton Theorem

Statement: Every Square Matrix satisfies its characteristic equation.

If A is given square matrix of order n , λ is an eigenvalue of A and I is an identity matrix of order n .

Then its characteristic equation is given by

$$|A - \lambda I| = 0.$$

$$a_n\lambda^n + a_{n-1}\lambda^{n-1} + a_{n-2}\lambda^{n-2} + \dots + a_2\lambda^2 + a_1\lambda + a_0 = 0$$

then by Cayley-Hamilton Theorem,

$$a_nA^n + a_{n-1}A^{n-1} + a_{n-2}A^{n-2} + \dots + a_2A^2 + a_1A + a_0 = 0.$$

Ex. Verify Cayley-Hamilton Theorem for the matrix A, hence find

$$A^{-1} \text{ \& } A^4. A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

Soln. The characteristic equation is given by $|A - \lambda I| = 0$

$$\therefore \begin{vmatrix} 2-\lambda & 0 & -1 \\ 0 & 2-\lambda & 0 \\ -1 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$\text{Using : } -\lambda^3 + S_1\lambda^2 - S_2\lambda + |A| = 0$$

Where $S_1 = \text{Trace } A$

$S_2 = \text{Sum of Minors of Diagonal Elements}$

$|A| = \text{Determinant of } A$

For given Matrix A,

$$S_1 = 6, S_2 = 11, |A| = 6$$

$$\therefore f(\lambda) = \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0 : \text{ The characteristic equation}$$

By Cayley-Hamilton Theorem, A Should satisfy the characteristic equation.

Verification:

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & 0 & -4 \\ 0 & 4 & 0 \\ -4 & 0 & 5 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 14 & 0 & -13 \\ 0 & 8 & 0 \\ -13 & 0 & 14 \end{bmatrix}$$

$$\begin{aligned} \therefore f(A) &= A^3 - 6A^2 + 11A - 6I \\ &= \begin{bmatrix} 14 & 0 & -13 \\ 0 & 8 & 0 \\ -13 & 0 & 14 \end{bmatrix} - 6 \begin{bmatrix} 5 & 0 & -4 \\ 0 & 4 & 0 \\ -4 & 0 & 5 \end{bmatrix} + 11 \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix} - 6I \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\text{Hence, } \therefore f(A) = A^3 - 6A^2 + 11A - 6I = 0 \dots\dots\dots(1)$$

ie. A satisfies its characteristic equation. Hence Cayley-Hamilton theorem is verified.

To find A^{-1} ,

Pre-multiplying (1) by A^{-1}

$$\therefore A^2 - 6A + 11I - 6A^{-1} = 0$$

$$\therefore A^{-1} = \frac{1}{6} \begin{bmatrix} 4 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}$$

To find A^4 ,

Pre-multiplying (1) by A

$$\therefore A^4 - 6A^3 + 11A^2 - 6A = 0$$

$$\therefore A^4 = \begin{bmatrix} 41 & 0 & -40 \\ 0 & 16 & 0 \\ -40 & 0 & 41 \end{bmatrix}$$

Ex. Use Cayley Hamilton theorem to find $A^7 - 9A^2 + I$.

Where $A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$.

The characteristic equation is given by $|A - \lambda I| = 0$

$$\therefore \begin{vmatrix} 1-\lambda & 4 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$\therefore (1-\lambda)^2 - 4 = 0$$

$$\therefore f(\lambda) = \lambda^2 - 2\lambda - 3 = 0 : \text{The characteristic equation} \\ \lambda = -1, 3$$

By Cayley-Hamilton Theorem, A satisfies the characteristic equation.

$$\therefore f(A) = A^2 - 2A - 3I = 0$$

$$\text{Let } g(A) = A^7 - 9A^2 + I$$

As coefficients of $f(A)$ and $g(A)$ are not same/similar, we use division algorithm

$$g(A) = f(A) \cdot q(A) + r(A); \text{ where degree of } r(A) < \text{degree of } f(A).$$

$$\therefore g(A) = 0 \cdot q(A) + a_0A + a_1I$$

$$A^7 - 9A^2 + I = a_0A + a_1I \dots\dots\dots 1$$

Eigenvalues of A satisfies this equation.

$$\therefore \lambda^7 - 9\lambda^2 + I = a_0\lambda + a_1I$$

$$\text{For } \lambda = -1$$

$$-1 - 9 + 1 = -9 = -a_0 + a_1 \dots\dots\dots 2$$

$$\text{For } \lambda = 3, \quad 2187 - 9(9) + 1 = 2107 = 3a_0 + a_1 \dots\dots\dots 3$$

Solving (2) & (3),

$$\text{We get, } a_0 = 529 \quad \& \quad a_1 = 520$$

By (1),

$$g(A) = A^7 - 9A^2 + I = 529A + 520I$$

Example 6 : Find the characteristic equation of the matrix A given below and hence, find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$, where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

Sol. : The characteristic equation is

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$\therefore (2-\lambda)[(1-\lambda)(2-\lambda)-0] - 1[0-0] + 1[0-1(1-\lambda)] = 0$$

$$\therefore (4-4\lambda+\lambda^2)(1-\lambda) - (1-\lambda) = 0$$

$$\therefore 4-4\lambda+\lambda^2-4\lambda+4\lambda^2-\lambda^3-1+\lambda = 0$$

$$\therefore \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0.$$

This equation is satisfied by A .

Now dividing $\lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1$ by $\lambda^3 - 5\lambda^2 + 7\lambda - 3$,

$$\begin{array}{r} \lambda^3 - 5\lambda^2 + 7\lambda - 3 \overline{) \lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1} \\ \underline{\lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5} \\ \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 \\ \underline{\lambda^4 - 5\lambda^3 + 7\lambda^2 - 3\lambda} \\ \lambda^2 + \lambda + 1 \end{array}$$

$\Rightarrow$ In terms of matrix A , we have

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$$

$$= (A^5 + A)(A^3 - 5A^2 + 7A - 3I) + (A^2 + A + I)$$

$$= A^2 + A + I$$

$\therefore A^3 - 5A^2 + 7A - 3I = 0$ by C.I. Theorem

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Eigen Values of matrix

Roots of Characteristic Equation of a Square Matrix are called characteristic roots / latent roots / characteristic values / eigen values / proper values of the matrix.

Eigen Vectors of matrix

If λ_1 is one of the eigen values of the matrix A , then Eigen Vector X corresponding to λ_1 is given by $[A - \lambda_1 I]X = 0$

Note

- ❖ Only square matrix possess eigenvalues.
- ❖ A square matrix of order n will have at the most n eigenvalues.
- ❖ Matrix of order n will have exactly n numbers of eigenvalues (may be distinct or repeated)
- ❖ Matrix may have real or complex eigenvalues.
- ❖ Eigenvalues may be equal to 0.
- ❖ Eigenvalues of a diagonal matrix and triangular matrix are its diagonal elements.
- ❖ The Sum of Eigenvalues = Trace of Matrix
- ❖ Product of the eigenvalues = determinant of matrix
- ❖ If a matrix A is singular then one of the eigenvalue of A must be zero.
- ❖ A and A^T have the same characteristic polynomial and hence their eigen values are identical
- ❖ If A is a square matrix and λ is an eigenvalue of A . Then (i) $\alpha\lambda$ is an eigenvalue of αA . (ii) Then for any integer $s \geq 0$, λ^s is an eigenvalue of A^s . (iii) $q(\lambda)$ is an eigenvalue of the matrix $q(A)$. (iv) λ^{-1} is an eigenvalue of A^{-1} . (v) $\frac{|A|}{\lambda}$ is an eigenvalue of $\text{adj}(A)$
- ❖ Eigenvalues of a unitary matrix are of unit modulus. ($\lambda\bar{\lambda} = 1$)
- ❖ Eigenvalues of a orthogonal matrix are of unit modulus.
- ❖ Hermitian Matrices have real eigen values
- ❖ Eigenvalues of Skew Hermitian Matrices and of real Skew symmetric matrix are either purely imaginary or 0.
- ❖ Eigenvectors are non-zero column vectors. Eigenvectors are Linearly independent.
- ❖ If X is eigenvector of A corresponding to some eigenvalue λ , then any non-zero multiple of X is also an eigenvector for same λ .
- ❖ If X is eigenvector of A corresponding to some eigenvalue λ , then eigenvector of any polynomial in $f(A)$ is X corresponding to some eigenvalue $f(\lambda)$.

Calculate Eigenvalues & Eigenvectors

Steps to be followed: A-square matrix of order n, I identity matrix of order n, λ any scalar (eigenvalue to be determined), X column vector of order n (eigenvector to be determined)

- **Find Characteristic Matrix:** $A - \lambda I$
- **Find Characteristic equation:** $|A - \lambda I| = 0$
- **Solve Characteristic equation and find its roots.**
(roots are called eigenvalues)
- **For each eigenvalue,** solve $[A - \lambda I] X = 0$ to determine nonzero column vector X.

(NOTE: If eigen values are distinct, we can use any one method to find eigen vectors –(i) row echelon method /(ii) algebraic method/(iii) Cramer's rule. For repeated eigen values we can use row echelon method only.)

Example 2 : Find the eigenvalues and eigenvectors of the following matrix.

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} 1-1 & -1 & 1 \\ -1 & 3-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{bmatrix}$$

Sol. : The characteristic equation is

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 3-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\therefore (3-\lambda)[(3-\lambda)^2 - 1] + 1[-1(3-\lambda) + 1] + 1[1 - (3-\lambda)] = 0$$

$$\therefore \lambda^3 - 9\lambda^2 + 24\lambda - 20 = 0 \quad \therefore \lambda^3 - 2\lambda^2 - 7\lambda^2 + 14\lambda + 10\lambda - 20 = 0$$

$$\therefore (\lambda - 2)(\lambda^2 - 7\lambda + 10) = 0 \quad \therefore (\lambda - 2)(\lambda - 2)(\lambda - 5) = 0$$

$$\therefore \lambda = 2, 2, 5$$

(i) For $\lambda = 2$, $[A - \lambda_1 I] X = O$ gives

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore x_1 - x_2 + x_3 = 0$$

The rank of the matrix is 1 and the number of variables is 3. Hence, there are $3 - 1 = 2$ linearly independent solutions.

Putting $x_2 = s$, $x_3 = -t$, we get $x_1 = x_2 - x_3 = s + t$.

$$\therefore X = \begin{bmatrix} s+t \\ s \\ 0-t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Now, the vectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ are linearly independent. (Verify it.)

Hence, corresponding to $\lambda = 2$, the eigen vectors are $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$.

(ii) For $\lambda = 5$, $[A - \lambda_2 I] X = O$ gives

$$\begin{bmatrix} -2 & -1 & 1 \\ -1 & -2 & -1 \\ 1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{By } \begin{matrix} R_2 + R_1 \\ R_3 + 2R_1 \end{matrix} \begin{bmatrix} -2 & -1 & 1 \\ -3 & -3 & 0 \\ -3 & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{By } \begin{matrix} R_3 - R_2 \\ -\frac{1}{3}R_2 \end{matrix} \begin{bmatrix} -2 & -1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore -2x_1 - x_2 + x_3 = 0; \quad x_1 + x_2 = 0$$

Putting $x_2 = -t$, we get $x_1 = t$ and $x_3 = 2x_1 + x_2 = 2t - t = t$.

$$\therefore X = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}. \quad \text{Hence, corresponding to } \lambda = 5, \text{ the eigenvectors is } \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Similarities of Matrices

- ❖ **Definition :** If A and B are two square matrices of order n then B is said to be similar to A if there exists a non-singular matrix M such that

$$B = M^{-1}AM$$

- ❖ **Definition :** A square matrix A is said to be **diagonalisable** if it is similar to a diagonal matrix.

Combining the two definitions we see that A is diagonalisable if there exists a matrix M such that $M^{-1}AM = D$ where D is a diagonal matrix. In this case M is said to **diagonalise A** or **transform A** to diagonal form.

If A is similar to B and B is similar to C, then A is similar to C.

If A and B are similar matrices then $|A| = |B|$

If A and B are two similar matrices then they have the same eigen values.

- ❖ **Algebraic Multiplicity (AM) of an eigen value.**

If λ_1 is an eigen value of the characteristic equation $|A - \lambda I| = 0$ repeated t times then t is called the algebraic multiplicity of λ_1 .

- ❖ **Geometric Multiplicity(GM) of an eigen value.**

If s is the number of linearly independent eigen vectors corresponding to the eigen value λ_1 then s is called the geometric multiplicity of λ_1 . So, the numbers of linearly independent solutions of $(A - \lambda_1 I)X = 0$ is s and the rank of the matrix $A - \lambda_1 I$ will be $n_1 - s$.

- ❖ **A Matrix is diagonalizable if AM=GM for all eigenvalues**

If $\lambda_1, \lambda_2, \lambda_3$ are eigen values of a matrix A of order 3 and X_1, X_2, X_3 are corresponding eigen vectors and matrix is diagonalizable, its diagonal matrix and transforming matrix or

diagonalizing matrices are given as $D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$ and $M = [X_1 \ X_2 \ X_3]$

Show that the matrix $A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$ is diagonalisable. Find the diagonal form D and the diagonalising matrix M .

Sol.: The characteristic equation is

$$\begin{vmatrix} -9 - \lambda & 4 & 4 \\ -8 & 3 - \lambda & 4 \\ -16 & 8 & 7 - \lambda \end{vmatrix} = 0$$

$$\therefore (1 + \lambda)(1 + \lambda)(3 - \lambda) = 0$$

$$\therefore \lambda = -1, -1, 3$$

(i) For $\lambda = -1$, $[A - \lambda_1 I] X = 0$ gives $\begin{bmatrix} -8 & 4 & 4 \\ -6 & 4 & 4 \\ -16 & 8 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

By $R_2 - R_1$ $\begin{bmatrix} 2 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $-(1/4)R_1$ $\begin{bmatrix} 2 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
 $\therefore 2x_1 - x_2 - x_3 = 0$

The rank of the coefficient matrix is $r = 1$. The number of unknowns is $n = 3$. Hence, there are $3 - 1 = 2$ linearly independent solutions.

Putting $x_2 = 2t$ and $x_3 = 2s$, we get

$$2x_1 = x_2 + x_3 = 2t + 2s \therefore x_1 = t + s$$

$$\therefore x_1 = \begin{bmatrix} s + t \\ 0 + 2t \\ 2s + 0 \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$\therefore$ Corresponding to the eigen values -1 , we get the following two linearly independent eigen vectors.

$$X_1 = [1, 0, 2]' \text{ and } X_2 = [1, 2, 0]'$$

(ii) For $\lambda = 3$, $[A - \lambda_2 I] X = 0$ gives

$$\begin{bmatrix} -12 & 4 & 4 \\ -8 & 0 & 4 \\ -16 & 8 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ By } R_2 - R_1 \begin{bmatrix} -12 & 4 & 4 \\ 4 & -4 & 0 \\ 4 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{By } R_3 + R_2 \begin{bmatrix} -12 & 4 & 4 \\ 4 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{By } -(1/4)R_1 \begin{bmatrix} 3 & -1 & -1 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore 3x_1 - x_2 - x_3 = 0 \text{ and } x_1 - x_2 = 0 \therefore x_1 = x_2.$$

Putting $x_2 = t$, we get $x_1 = x_2 = t$ and $x_3 = 3x_1 - x_2 = 3t - t = 2t$.

$$\therefore X_3 = \begin{bmatrix} t \\ t \\ 2t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \therefore X_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Here the geometric multiplicity of each eigen value of A is equal to its algebraic multiplicity, A is diagonalisable.

Since, $M^{-1}AM = D$, the matrix $A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$ will be diagonalised

$$\text{to the diagonal matrix } D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\text{by the transforming matrix } M = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 2 & 0 & 2 \end{bmatrix}$$

Functions of Square Matrices

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, find A^{50} .

Sol. : The characteristic equation is

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 1 & 0-\lambda & 1 \\ 0 & 1 & 0-\lambda \end{vmatrix} = 0$$

$$\therefore (1-\lambda) [(-\lambda)(-\lambda) - 1] = 0$$

$$\therefore (1-\lambda)(\lambda^2 - 1) = 0$$

$$\therefore \lambda = 1, 1, -1.$$

Since, the matrix is of order 3, we consider

$$\Phi(A) = A^{50} = \alpha_2 A^2 + \alpha_1 A + \alpha_0 I \quad \dots(1)$$

X satisfies this equation.

$$\therefore \lambda^{50} = \alpha_2 \lambda + \alpha_1 \lambda + \alpha_0 \quad \dots(2)$$

Putting $\lambda = 1, \lambda = -1$, we get

$$1 = \alpha_2 + \alpha_1 + \alpha_0 \quad \dots(3)$$

$$1 = \alpha_2 - \alpha_1 + \alpha_0 \quad \dots(4)$$

Differentiating (2), w.r.t. X, we get

$$50\lambda^{49} = 2\alpha_2 \lambda + \alpha_1$$

Putting $\lambda = 1$, we get

$$50 = 2\alpha_2 + \alpha_1 \quad \dots(5)$$

Solving (iii), (iv) and (v), we get

$$\alpha_2 = 25, \alpha_1 = 0 \text{ and } \alpha_0 = -24.$$

Putting these values in (1), we get $A^{50} = 25A^2 - 24I$

$$\text{But } A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\therefore A^{50} = 25 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} - 24 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$$